

VIRAJ THAKUR

Python Developer — Backend Development · REST APIs · Async Systems

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PROFESSIONAL SUMMARY

Final-year Computer Engineering graduate specializing in Python backend development. Built and tested a role-based FastAPI service (Finance_backend) enforcing three-tier access control across Admin, Analyst, and Viewer roles, backed by a fully passing pytest suite. Designed an asynchronous SaaS platform (LieLens) using Django, Celery, and Redis to run NLP-based scoring without blocking user requests. Comfortable across the backend lifecycle — schema design, authentication, containerization, and automated testing.

TECHNICAL SKILLS

Languages	Python (primary) · SQL · JavaScript
Backend	FastAPI · Django · Flask · REST API Design · SQLAlchemy 2.0 · Pydantic v2 · JWT Authentication · RBAC
NLP / ML	Scikit-learn · Logistic Regression · Feature Engineering · LLMs · GenAI · RAG · LangChain
Databases	PostgreSQL · MySQL · Redis (Pub/Sub, Caching) · SQLite
Testing	Pytest · httpx · Automated Test Suites
DevOps	Docker · Docker Compose · GitHub Actions (CI) · Git
Concepts	OOP · CRUD · Async Processing · SaaS Architecture · Dual-Database Design

PROJECTS

Finance_backend — Role-Based Financial API Platform

github.com/VirajThakur03/Finance_backend

- Secured a personal finance API by designing three-tier role-based access control (Admin, Analyst, Viewer), with correct enforcement verified by a fully passing automated pytest suite.
- Built asynchronous REST endpoints in FastAPI with SQLAlchemy 2.0 and Pydantic v2, supporting dual-database operation (PostgreSQL in production, SQLite for local development) so the app runs identically in both environments.
- Containerized the service with Docker and Docker Compose, reducing environment setup to a single command for both local development and deployment.

Stack: FastAPI · SQLAlchemy 2.0 · Pydantic v2 · JWT · Pytest · PostgreSQL / SQLite · Docker

LieLens — Writing Risk Intelligence SaaS

github.com/VirajThakur03/lielens-ai-saas

- Designed an explainable writing-risk scoring system by combining rule-based NLP feature extraction with a Logistic Regression classifier, producing five interpretable sub-scores (confidence, exaggeration, credibility, emotional intensity, final risk) instead of a single opaque output.
- Moved heavy NLP analysis off the request thread by building an async processing pipeline with Celery and Redis, keeping the API responsive during scoring; Celery Beat automates monthly usage resets and insight emails.
- Scaffolded SaaS infrastructure — Stripe webhook handling, API-key issuance, and plan-based rate limiting — applying the same JWT/RBAC access patterns used in Finance_backend.

Stack: Django · Celery · Redis · PostgreSQL · Scikit-learn · Docker · JWT · Stripe (webhook scaffold)

Skillo — Local Skill Marketplace

github.com/VirajThakur03/skillo

- Built a two-role marketplace (seeker and provider) with JWT role-based authentication and REST endpoints for listing and booking local skills.
- Added real-time booking notifications with Socket.IO and Redis pub/sub, and containerized the full stack (Flask, PostgreSQL, Redis) with Docker Compose for single-command, reproducible setup.

Stack: Flask · PostgreSQL · Redis · Socket.IO · JWT · Docker

Speech Emotion Recognition

github.com/VirajThakur03/Speech-Emotion-Recognition-

- Built an end-to-end ML pipeline classifying 6 emotions from raw speech on the RAVDESS dataset by engineering 40-dimensional MFCC features and tuning an SVM classifier, reaching 82% accuracy.
- Improved baseline accuracy by ~12 percentage points through iterative preprocessing, evaluated with precision, recall, and F1-score.

Stack: Python · Scikit-learn · Librosa · Pandas · NumPy · Matplotlib · Seaborn

EDUCATION

B.E. Computer Engineering — Sindhudurg Shiksha Prasarak Mandal (CGPA: 8.20)

2021 – 2025

B.Sc. Mathematics — Balasaheb Khardekar College, Vengurla (81.67%)

2020 – 2021

CERTIFICATIONS

- Programming in Python — Besant Technologies (Feb 2024)
- Generative AI & Model Lifecycle — NXT Wave (Dec 2024)
- SQL — DDL / DML / Joins / Aggregations — NXT Wave (Sep 2024)
- AWS Solutions Architecture Job Simulation — Forage (Oct 2024)